INDOORS DEMO ASSET SCHEMA

For the exact dataset, field-population, domain, relationship, contingent-value, and attribute-rule delta between the original medical Indoors geodatabase and the generated single-facility output, see [Vertex Assets Geodatabase Delta](vertex_assets_gdb_delta.md).

Schema relationships
Facilities -> Levels -> Units
Units -> Indoor Asset Location -> Asset Lifecycle
Facilities -> Exterior Asset Points
Asset Lifecycle -> Inspections, Repairs, Work Orders, External References, Field Work, Media

Facilities, Levels, and Units remain ArcGIS Indoors-owned source records. The generator creates the AssetManagement feature dataset and its asset-management records in the copied output geodatabase.

INSPECTION WORKFLOW

Inspect Indoors Geodatabase produces a versioned JSON decision record for a selected source geodatabase and facility profile. Its inspection section is intended for review: compatibility, Unit and room-use counts, mapped and unmapped USE_TYPE values, and a recommended action. Full field, domain, relationship, and imagery discovery remains available in technical_details.

Generate Demo Assets can consume this report when the report source path, selected profile name, and profile version match its inputs. It uses the report to make profile coverage visible before it writes. It always repeats minimum live-source validation immediately before copying the geodatabase, because an inspection report cannot prove that the GDB has not changed.

Live-source validation requires Sites, Facilities, Levels, and Units, together with their business-key parent fields (SITE_ID, FACILITY_ID, LEVEL_ID, and UNIT_ID) and Units.USE_TYPE. This prevents a source from passing preflight when the hierarchy needed for generation is incomplete.

INTEGRATION WORKFLOW

Integrate Generated Assets combines generated records with a target Indoors workspace without modifying either input. It creates a new output geodatabase by copying the target and then recreates the generator-owned AssetManagement schema in that output.

Before it writes, the tool verifies that every SourceUnitGlobalID in generated indoor assets exists in the target Units class, and that every SourceFacilityGlobalID in generated exterior assets exists in the target Facilities class. It rejects a missing

parent reference instead of creating orphaned records.

File geodatabases create new GlobalIDs during record insertion. The tool therefore remaps generated location and lifecycle GlobalIDs in the new output and updates all dependent lifecycle foreign keys before validating the completed relationships. This preserves the target Indoors hierarchy and leaves the generated and target input geodatabases unchanged.

GlobalIDs remain authoritative relationship keys, but they are not stable generation inputs because a copied or rebuilt file geodatabase assigns new values. Deterministic allocation instead uses business identities: Facility_ID|Level_ID|Unit_ID for indoor assets and Facility_ID for exterior assets. Rebuilding the same facility with the same profile, hierarchy, and seed therefore preserves generated asset IDs, quantities, and allocation. Relationship rows are remapped to the current output GlobalIDs, so consumers must not use old GlobalIDs as cross-build identifiers.

Input | Action | Result
Generated demo GDB | Read-only source of generated records | Remains unchanged
Target Indoors GDB | Copied before changes | Remains unchanged
New integrated GDB | Receives copied target and remapped generated records | Standalone integrated workspace

FACILITY PROFILES

The tool selects assets through a versioned facility profile. A profile maps Units.USE_TYPE values to explicit templates; it does not infer a room type. Every template controls its asset type, asset class, quantity range, criticality, maintenance interval, and system dependencies.

Profile | Intended coverage | Asset examples
Hospital | General hospital Indoors data with core clinical and building spaces | hospital beds, patient monitors, operating table
Vertex | Enhanced Hospital profile for the supplied Vertex medical facility; covers all discovered ‘USE_TYPE‘ values | all Hospi
Government Office | General office and public-service spaces | office workstations, security equipment, IT/network hardware, b

Vertex includes generic, common clinical assets where the source has a clinical use without a specific equipment inventory. For example, pre-op and post-op rooms receive preparation/recovery furniture and patient-monitoring assets; treatment, pediatric, psychiatric, outpatient, and infusion spaces receive treatment chairs, vital-signs monitors, and clinical handwash sinks. These are pseudo-assets for demonstration, not an asserted facility equipment inventory.

VERTEX SOURCE COVERAGE

The Vertex profile explicitly maps the supplied source uses into these template groups:

Source-use group | Vertex template group | Typical generated assets

Pre-Op, Post-Op, Endoscopy | Preparation, recovery, procedure | stretchers, recovery beds, procedure tables, monitors
Cardiology, Pediatrics, Infusion, Outpatient, Psychiatric | Clinical specialty or treatment | treatment chairs, diagnostic workstatio
Laboratory, Pharmacy, Rehabilitation | Department-specific clinical support | lab analyzers, medication refrigerators, therapy eq
Office, Doctor Office, HR, Billing, Mail, Copy, Phone | Office or clinical office | workstations, office furniture, wall diagnostic sets
Admission, Entry Lobby | Reception | desks, workstations, access-control readers
Meeting, Training | Meeting room | video conferencing systems, presentation displays
Kitchen, Nutrition, Locker Room, Lounge, Break Room | Staff or food service | lockers, commercial refrigerators, handwash sin
Restroom, Storage, Medical Records | Support services | plumbing fixtures, shelving, records storage/workstations
Elevator, Parking Garage, Outdoor Space, Open to Below | Transport or common area | elevator control panels, parking lightin
Fire Closet, Mechanical, Electrical, Telecom, Janitorial, Stairway | Building infrastructure | fire panels, air handling units, switch

INTEGRATION BRIDGES

AssetExternalReference is an open, vendor-neutral mapping table. A single asset can have zero or many rows: for example, one Salesforce Asset reference and one IBM Maximo ASSETNUM reference. SystemName and SyncStatus are coded domains; ExternalObjectType, ExternalRecordID, and RecordUrl remain intentionally open to support other enterprise asset/resource-management systems. Store endpoint URLs, OAuth tokens, passwords, and other credentials outside the geodatabase.

AssetFieldWorkLink bridges an asset to ArcGIS field workflows. It supports either ArcGIS Inspection or Survey123, identifies the target portal item and layer/form, and can store the related inspection/Survey123 GlobalID once a response exists. The application integration is deferred: these records are safe link contracts, not a claim that the toolbox publishes forms or calls ArcGIS Online.

AssetMediaReference is a one-to-many link from an asset to OrientedImage, Panorama360, Photo, or Document records. It stores a reference to an ArcGIS Oriented Imagery Catalog item or a secured external image, together with optional capture and camera-orientation metadata. Images themselves and credentials are not stored in the geodatabase.

CSV SOURCE ALIGNMENT

The generated AssetLifecycle and location records mirror the supplied Asset_Data_from_AMS.csv, Critical_Indoor_Assets.csv, and Medical_Equipment.csv contract: Asset_ID, category/subtype fields, make/model/serial, owner, maintainable/fixed flags, criticality rating, Unit_ID, Level_ID, Facility_ID, Site_ID, barcode, photo URL, and height. GlobalIDs remain the authoritative geodatabase relationship keys; business keys remain available for straightforward import/export.

AssetInspection, AssetRepair, and AssetWorkOrder mirror the supplied Inspections.csv, Repairs.csv, and Work_Orders.csv fields. Each row has both AssetLifecycleGlobalID and AssetPointGlobalID. The first preserves lifecycle-centered analysis; the second directly relates the row to IndoorAssetLocation, making history available from asset-layer popups without traversing the intermediate lifecycle table.

When publishing, include IndoorAssetLocation, all three history tables, and their relationship classes in the same feature service. Configure related-record elements in the web map or application popup after publishing; the geodatabase relationships supply the association but do not automatically design the popup.

Generated Unit_ID, Level_ID, Facility_ID, and Site_ID text fields inherit their lengths from the corresponding source Indoors hierarchy fields when available. In particular, AssetManagement/IndoorAssetLocation.Level_ID must match Indoors/Levels.LEVEL_ID for ArcGIS Pro to configure the asset layer as floor aware without a field-length warning.

ENRICHED ASSET PROFILE

The Vertex enrichment workflow adds deterministic planning fields to AssetLifecycle. These values are synthetic and internally consistent; they are intended for dashboard filtering, replacement planning, warranty analysis, and prioritization rather than as claims about real equipment.

Field group | Fields | Contract
Lifecycle | ‘LifecycleStage‘, ‘ExpectedLifeYears‘, ‘AssetAgeYears‘, ‘RemainingLifeYears‘, ‘ReplacementDueDate‘, ‘Replacemer
Inspection/maintenance | ‘LastInspectionDate‘, ‘OverdueDays‘, ‘MaintenanceIntervalDays‘, ‘AnnualMaintenanceCost‘ | Latest i
Financial | ‘EstimatedReplacementCost‘, ‘PurchaseDate‘, ‘PurchaseCost‘ | Purchase precedes commissioning and historical pu
Procurement/warranty | ‘Vendor‘, ‘WarrantyExpirationDate‘, ‘WarrantyStatus‘ | Vendors and warranty terms are class-specific; s
Risk | ‘RiskScore‘, ‘RiskLevel‘ | Score is 0-100 and combines criticality, condition, lifecycle stage, and maintenance lateness; le

INTEGRITY CONTROLS

Dataset | Controlled fields | Integrity mechanism
‘AssetLifecycle‘ | ‘AssetClass‘, ‘Condition‘, commissioning and maintenance dates | coded domains, AssetClass/Condition cont
‘ExteriorAssetPoints‘ | ‘PlacementType‘, ‘Condition‘ | coded domains
‘AssetExternalReference‘ | ‘SystemName‘, ‘SyncStatus‘ | coded domains; indexed AssetGlobalID and external record ID
‘AssetFieldWorkLink‘ | ‘WorkflowType‘, ‘Status‘ | coded domains; indexed AssetGlobalID
‘AssetInspection‘ | ‘Condition_Rating‘, ‘Repair_Needed‘ | coded domains; indexed lifecycle and asset-point GlobalIDs
‘AssetRepair‘ | parent references | indexed lifecycle and asset-point GlobalIDs
‘AssetWorkOrder‘ | ‘Status‘ | coded domain; indexed lifecycle and asset-point GlobalIDs
‘AssetMediaReference‘ | ‘MediaType‘ | coded domain; indexed AssetLifecycleGlobalID

The generator places attached assets on the related Facility boundary. Exterior assets use a scale-relative outward buffer from that boundary, so they remain outside the Facility polygon in either projected or geographic coordinate systems.